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NEW AGE TOOLING BOF @ NETDEV 0X1A

The Cost Asymmetry of AI-Generated Code a Case Study in the SRv6 Subsystem

Andrea Mayer¹, Stefano Salsano^{1,2}

¹University of Rome Tor Vergata, Italy · ²CNIT, Italy

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We are not here to blame anyone — we are here to learn, and to improve the kernel development process.

The case: one patchset, everything green

[PATCH v2 0/7] seg6: add SRv6 Mobile User Plane (RFC 9433)

- **6 new behaviors** + documentation, in one series
- **5,155 insertions**, 12 files — seg6_local.c grows 2.7k → 5k lines
- **v1 → v2 in one day**, CI findings all addressed
- **2,402 lines of selftests** (47% of the series) — all green
- Textbook commit messages, precise RFC citations

Takeaway. Every cheap quality signal looked good. None of them measured correctness.

0/7 cover letter

1/7 End.MAP (+191)

2/7 End.M.GTP4.E (+1,232)

3/7 End.M.GTP6.E (+717)

4/7 End.M.GTP6.D (+1,013)

5/7 End.M.GTP6.D.Di (+652)

6/7 H.M.GTP4.D

7/7 Documentation

The asymmetry, measured in the wild

1 **Generation: ~1 day per iteration.**

5,155 lines, six behaviors; v1 on May 4, v2 on May 5.

2 **Expert review: one month.**

8 emails, ~40 findings (May 8 – June 8) — and it covered only 5 of 7 patches.

Outcome: start over.

3 Nothing merged; full rework agreed — and the maintainer walked away with *more* work (two prep workstreams).

Takeaway. A 1 : 5–10 generation-to-review cost ratio is not a thought experiment — this thread measured it.

Anatomy of ~40 review findings

Class	What we found	Why it hurts
Spec conformance	Patch 4 implements neither RFC 9433 §6.3 nor §6.4; RFC 6040 mis-cited	Fatal, invisible to CI
Security / validation	Malformed SRH accepted as absent → HMAC bypass; HMAC invalid by construction	Silent
UAPI design	Required-but-unused attributes; NH6 / SRH / OIF semantics overloaded	Irreversible once merged
Data path & structure	GSO broken, stale checksums, unchecked fragments; ×5 duplicated logic	Bugs hide in the copies

Takeaway. The most damaging classes — spec, security, UAPI — are exactly the ones no automated pipeline caught.

The selftest that passed by coincidence

The bug.

- 1 End.MAP rewrites the IPv6 DA — the ICMPv6 checksum covers the DA, so it must be updated. It isn't.

The mystery.

The AI reviewer flagged it — yet the generated selftest stayed green.

The resolution.

- 3 The two test addresses have equal 16-bit word sums: `2001:db8:f::1` vs `2001:db8:2::e` — `0x000f + 0x0001 = 0x0002 + 0x000e`. Change one address → `Icmp6InCsumErrors`.

"The AI bot was right, but when I ran the selftest it passed." — the human reviewer, digging in

Takeaway. Generated tests share the generator's blind spots — green CI $\neq$ correct.

"Plausible-but-wrong": the signature failure mode

The code looks idiomatic. The prose is precise. The substance is wrong:

- Comments cite RFC 9433 **step numbers** — the code implements **neither §6.3 nor §6.4**
- Comments describe **code that is not there** (hop-limit decrement, LOCAL_OUT claim)
- RFC 6040 cited **for something it does not cover**
- The commit message's own example **never uses a required attribute**
- Commit message lists a drop reason **the diff never adds**

Punchline. The AI reviewer scored this very patch cleanest of the series: **0 critical · 0 high · 0 medium · 1 low.**

Takeaway. Surface signals of quality are decoupled from correctness — checking code against the *intent* of a spec is still human work.

Human vs AI review, on the same patches

Same perimeter (patches 1–5) — AI reviewer: Sashiko (Patchwork), every finding re-verified against kernel source.

	Human (5 weeks)	AI (hours, ~\$34 in tokens)
Outcome-deciding classes (spec · UAPI · architecture)	all of them	zero
Overlap	~40 findings	reproduced 9 of them (~22%)
Unique findings	~31, incl. every blocker	5 → 2 real but minor + 3 false positives
False positives	0	3 — each on an invented kernel mechanism, incl. its only Critical

One human sentence on hop-limit handling (*"the forwarding path handles it"*) pre-refuted two AI attack scenarios.

Takeaway. AI review re-enacts the asymmetry one level up: findings are cheap to generate, expensive to verify — here, verification *debunked more than it confirmed*.

Lessons: dos & don'ts

- **Disclose AI use** — an honesty norm, orthogonal to acceptance
- **Small series** — one behavior per patchset (the End.DT4/DT6/DT46 model)
- **UAPI is forever** — design it by hand, review it first
- **Don't trust generated tests** — they share the generator's blind spots
- **AI-review output = claims to verify**, not findings
- **Humans keep the merge decision** — and the spec, UAPI, architecture calls
- **Cheap guards work** — the early RFC-downgrade cost nothing and helped

Takeaway. The scarce resource is *expert maintainer attention* — every rule above exists to protect it.

Two proposals for this BoF

1. Improve AI review — the margins are wide

- **Feedback loop:** track "% wrong" next to "% unique" — every debunked finding becomes eval signal
- **Maintainer-owned, per-subsystem review prompts** — the mechanism already exists; the contribution pipeline doesn't

2. Desk rejection — on quality, not provenance

- Reject code because it's **badly written**, not because it's AI-written
- Mechanical, provenance-neutral gates: **clone density**, dead-code & cast fingerprints, commit-msg ↔ diff consistency
- Below the bar: **reject wholesale**, return the cost to the sender

Take-home. Price the review *before* it starts — no AI-detector arms race needed.

Thank you. Discussion: who defines the bar — CI, Patchwork, MAINTAINERS?

`andrea.mayer@uniroma2.it` · `stefano.salsano@uniroma2.it`